

Udit Agarwal

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EDUCATION

Indiana University Bloomington

Master of Science in Computer Science, GPA: 3.84/4.0

Bloomington, Indiana, US

Aug 2024 - May 2026

Kalinga Institute of Industrial Technology (KIIT University)

Bachelor of Technology in Computer Science and Engineering, GPA: 8.85/10.0

Bhubaneswar, Odisha, India

Aug 2020 - May 2024

EXPERIENCE

Global Health Impact Project

Feb 2026 - Present

Software Engineer

Indiana University, Bloomington, Indiana, USA

- Developed a data-driven global health platform using React, TypeScript, and Python to analyze pharmaceutical impact across global populations
- Built scalable full-stack features and REST APIs to support forecasting and analytics workflows
- Contributed to a system modeling treatment coverage and efficacy to evaluate global health outcomes at scale
- Collaborated with cross-functional teams to translate research requirements into production-ready solutions

University Information Technology Services (UITS)

Aug 2025 - Present

Consultant - IT Services

Indiana University, Bloomington, Indiana, USA

- Diagnosed and resolved 100+ system issues weekly using structured troubleshooting and root-cause analysis to improve operational efficiency
- Managed identity and access systems, ensuring secure authentication and seamless user access
- Applied structured problem-solving to identify root causes and reduce recurring technical issues across enterprise systems
- Documented support workflows and system issues in ServiceNow to improve service delivery and team efficiency

PROJECTS

AI Mock Interview Platform - Next.js, Firebase, Node.js, Vapi AI, Gemini API

- Built a full-stack AI platform to simulate real-time interview scenarios using voice agents and LLM APIs, serving end-to-end user workflows
- Designed scalable backend APIs and integrated external AI services for dynamic question generation and contextual feedback
- Developed responsive UI and data pipelines for seamless user interaction, performance tracking, and structured insights

Retail Analytics Platform - Next.js, PostgreSQL, Drizzle ORM, Chart.js

- Designed a normalized PostgreSQL schema for 10000+ records with SQL-based ETL pipelines ensuring data integrity and consistency
- Built backend APIs and analytical queries to power dashboards and generate actionable business insights
- Developed interactive dashboards to visualize KPIs, business performance metrics, and trends across datasets

Brain Tumor Classification - Python, TensorFlow, Convolutional Neural Networks

- Developed deep learning models achieving 99.84 percent accuracy across 7000+ MRI scans using CNN-based architectures
- Benchmarked architectures including CNN, VGG16, InceptionV3, and EfficientNet to optimize performance and inference efficiency
- Co-authored IEEE publication on medical image classification using deep neural networks

TECHNICAL SKILLS

Languages: Python, JavaScript, TypeScript, SQL, C++

Frontend: React, Next.js, HTML, CSS, Tailwind CSS

Backend: Node.js, Express.js, FastAPI, REST APIs

Data and Analytics: Pandas, NumPy, Tableau, ETL Pipelines, Data Modeling, Statistical Analysis

Databases: PostgreSQL, MySQL, MongoDB, Redis

ML and AI: TensorFlow, Scikit-learn, Transformers, Hugging Face

Cloud and DevOps: AWS, Docker, CI/CD, Git

Systems and Concepts: System Design, API Design, Distributed Systems, Authentication, Data Storytelling